State-space Gromov–Hausdorff convergence of truncated tensor-product Camporesi–Higuchi spectral triples on $S^3 \times S^3$

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Abstract

We prove that the Connes–van Suijlekom truncations of the tensor-product Camporesi–Higuchi spectral triple $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ on $S^3 \times S^3$ at two (possibly distinct) focal lengths $\lambda_a, \lambda_b > 0$ converge to the continuum tensor-product triple in van Suijlekom's state-space Gromov–Hausdorff distance, with explicit rate $\Lambda(\mathcal{T}_{n_{\max,a}} \otimes \mathcal{T}_{n_{\max,b}}, \mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}) \leq \gamma_\lambda^{(ab)} \leq \lambda_a \gamma_{n_{\max,a}} + \lambda_b \gamma_{n_{\max,b}}$, of *leading constant* 1, where $\gamma_\lambda^{(ab)}$ is the joint first moment of the product Fejér measure and $\gamma_n = (4/\pi) \log n / n + O(1/n)$ inherits from the single-factor result of [20]. The proof is the lifted-state construction (Lemma 4.12), which uses neither a Berezin map nor any partial inverse: the dual reach reach_P is *unnecessary*, both almost-inverse defects being the same Fejér smoothing at the same moment. The theorem is [INTERNAL THEOREM] *in the translation-seminorm metrization* — the footing of Paper 38's `thm:main_unconditional`, with which both state spaces are metrized; that seminorm agrees with the truncated-Dirac commutator seminorm on the continuum and, unlike it, is non-degenerate at finite cutoff. The limit is the graded tensor product of the two odd (KO-3) Camporesi–Higuchi real spectral triples — Hilbert space $\mathcal{H}_a \otimes \mathbb{C}^2 \otimes \mathcal{H}_b$, Dirac $D_{\mathrm{CH}}^{(a)} \otimes \sigma_1 \otimes 1 + 1 \otimes \sigma_2 \otimes D_{\mathrm{CH}}^{(b)}$, grading $1 \otimes \sigma_3 \otimes 1$, real structure $J_{a,b} = J_a \otimes \sigma_1 K \otimes J_b$ at KO-dimension 6 (Lemma 4.13); the Clifford $\mathbb{C}^2$ is the doubling that carries the product grading and real structure, the naive $D_{\mathrm{CH}}^{(a)} \otimes I + \gamma_a \otimes D_{\mathrm{CH}}^{(b)}$ being unrealisable ($\gamma_a = 0$ on S^3). The strictly stronger Latrémolière propinquity is not claimed: it is not pursued here, and its dual-reach step would reintroduce the reach_P named gap of Papers 38/40 that the lifted-state route avoids; the object established is van Suijlekom state-space Gromov–Hausdorff convergence.

An earlier (B, P)-pair assembly (the retained Lemma 4.14) gave a weaker conditional bound with constant 2 *via the joint Lipschitz comparison constant $C_3^{(2)} \leq \sqrt{2}$; that route's height leg was refuted and its dual reach is a named gap, and it is superseded by the lifted-state route above.* The result extends [6, 15, 20] to the simplest non-abelian tensor case, and closes a tensor-product state-space-GH case that the published literature on products of metric spectral triples [1, 7, 8, 14] does not cover; existing tensor-product propinquity results require the second factor to be finite-dimensional, AF, abelian, or to reduce to a single-character / Hopf-S^1 structure.

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1 Introduction

The Connes–van Suijlekom [6] framework of truncated operator systems and the subsequent propinquity-convergence results of Hekkelman [9], Hekkelman–McDonald [10], Leimbach–van Suijlekom [15], and the present author [20] establish Gromov–Hausdorff convergence for spectral truncations of a *single* compact Riemannian spectral triple: for SU(2) with an explicit rate $(c+o(1))\log n_{\max}/n_{\max}$ [20], and for the flat torus qualitatively (Leimbach–van Suijlekom state no explicit rate). A natural and operationally important next question is whether the convergence behaviour transfers to *tensor products* of two such truncated triples.

The published propinquity tensor-product literature does not cover this case in its central form. Latrémolière [14] closes the *almost-commutative* case in which one factor is an infinite-dimensional Riemannian spectral triple and the other is finite-dimensional. Farsi–Latrémolière [7] prove spectral continuity for collapse of products of spectral triples in which at least one factor is abelian. Farsi–Latrémolière [8] treat continuity of a *single* metric spectral triple along an analytic path of Riemannian metrics — a setting distinct from the discrete-truncation product considered here. Aguilar [1] addresses tensor products $C(X)\otimes A$ where one factor is commutative and the other AF. Hekkelman–McDonald [10] treats a noncommutative integral on a single spectrally truncated triple, not a product. The published thread covers Riemannian-by-finite, Riemannian-by-AF, Riemannian-by-abelian, $C(X)$-by-AF, single-triple-along-paths, and single-truncation. None of these covers two infinite non-abelian Riemannian metric spectral triples.

The present paper closes that gap in the simplest physically natural non-Kähler case: two copies of the round $S^3 = \mathrm{SU}(2)$ Camporesi–Higuchi triple [4], each in its own focal-length scaling, tensored together via the Connes–Marcolli graded convention.

Let $\mathcal{T}^{\lambda}_{S^3} = (C^\infty(S^3), L^2(S^3,\Sigma), \lambda^{-1}D_{\mathrm{CH}})$ denote the round Camporesi–Higuchi spectral triple at *focal length* $\lambda > 0$, i.e. the unit-round triple with the Dirac operator rescaled so that the spectral gap is λ^{-1} rather than 1. (In the GeoVac hydrogenic context this corresponds to placing two electrons in distinct hydrogenic shells with focal lengths $\lambda_a = n_a/Z$ and $\lambda_b = n_b/Z$.) Form the tensor product

$$\mathcal{T}^{\lambda_a}_{S^3}\otimes\mathcal{T}^{\lambda_b}_{S^3} \;:=\; \big(C^\infty(S^3)\otimes C^\infty(S^3),\; L^2(S^3,\Sigma)\otimes L^2(S^3,\Sigma),\; D_{a,b}\big), \tag{1}$$

with Connes–Marcolli graded composed Dirac

$$D_{a,b} \;=\; \lambda_a^{-1}D^{(a)}_{\mathrm{CH}}\otimes I_b \;+\; \gamma_a\otimes\lambda_b^{-1}D^{(b)}_{\mathrm{CH}}, \tag{2}$$

where γ_a was described until 2026-09-04 as "the chirality grading on $L^2(S^3,\Sigma)$". **No such operator exists** [retracted 2026-09-04: p39-chirality-grading-on-s3]: a grading anticommuting with every generator of Cl(3) also anticommutes with the volume element $\omega = \sigma_1\sigma_2\sigma_3 = i\,I$, which in odd dimension is *central*, so $\gamma\omega = -\omega\gamma$ and $\gamma\omega = \omega\gamma$ together force $\gamma = 0$. Paper 38 states the same ("KO-3 carries no chirality", 2026-09-03) and Paper 32 has carried it since 2026-09-02; the correction did not reach this paper. The displayed $D_{a,b}$ is the *even*$\otimes$odd product formula, while this paper's KO arithmetic $3+3=6$ is the odd$\otimes$odd case, whose standard construction requires Clifford doubling ($\mathcal{H}_a\otimes\mathbb{C}^2\otimes\mathcal{H}_b$, $D = D^{(a)}_{\mathrm{CH}}\otimes\sigma_1\otimes 1 + 1\otimes\sigma_2\otimes D^{(b)}_{\mathrm{CH}}$). **That repair is named, not adopted**, and *whether Lemmas 4.6 and 4.10 hold for it is open* — they were proved for the operator displayed here. See the status note to Theorem 3.1. The combined object has KO-dimension $3+3=6 \pmod 8$. Truncate each factor independently at cutoffs $n_{\max,a}, n_{\max,b}$ and form the tensor-product truncated triple $\mathcal{T}_{n_{\max,a}}\otimes\mathcal{T}_{n_{\max,b}}$.

Theorem 1.1 (Main theorem; cf. §3)**.** *The truncated tensor-product triples $\mathcal{T}_{n_{\max,a}}\otimes\mathcal{T}_{n_{\max,b}}$ converge to $\mathcal{T}^{\lambda_a}_{S^3}\otimes\mathcal{T}^{\lambda_b}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance, with bound*

$$\Lambda\big(\mathcal{T}_{n_{\max,a}}\otimes\mathcal{T}_{n_{\max,b}}, \mathcal{T}^{\lambda_a}_{S^3}\otimes\mathcal{T}^{\lambda_b}_{S^3}\big) \;\le\; \gamma^{(ab)}_{\lambda} \;\le\; \lambda_a\,\gamma_{n_{\max,a}} + \lambda_b\,\gamma_{n_{\max,b}}, \tag{3}$$

of leading constant 1, *where* $\gamma_\lambda^{(ab)}$ *is the joint first moment of the product Fejér measure and* $\gamma_n = (4/\pi)\log n/n + O(1/n)$. *The convergence is [INTERNAL THEOREM] in the translation-seminorm metrization (Paper 38's footing); it is proved by the lifted-state construction (Lemma 4.12), for which the dual reach* reach_P *is unnecessary. The joint Lipschitz comparison constant* $C_3^{(2)} \leq \sqrt{2}$ *((19)) belongs to the* abandoned (B, P)*-pair route (Lemma 4.14) and is not a rate constant of this theorem.*

The proof is the lifted-state assembly (Lemma 4.12): the four facts (i)–(iv) of Paper 38's `thm:main_unconditional`, lifted factor-wise to $\mathrm{SU}(2) \times \mathrm{SU}(2)$ and metrized by the translation seminorm (Lemma 4.11), give a correspondence of distortion $\leq 2\gamma^{(ab)}$ and hence $d_{\mathrm{GH}} \leq \gamma^{(ab)} \leq \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}$; it uses neither a Berezin map nor a partial inverse, so reach_P never enters. The earlier (B, P)-pair route (Lemmas 4.6–4.14) is retained as the record of an abandoned approach, supplemented there by two ingredients specific to the product:

- a closed-form Lipschitz comparison estimate from the (tight) operator-norm triangle bound $\|A + B\|_{\mathrm{op}} \leq \|A\| + \|B\|$ on the two Connes–Marcolli Leibniz terms (Lemma 4.6); and
- an explicit Stein–Weiss-type bound on the joint cross-Stein–Weiss *panel* term, again via the triangle bound applied to the joint commutator (Lemma 4.14). This term was formerly assembled into the rate as a Lipschitz-distortion height; that step is withdrawn (2026-09-04) and the rate now rests on reach-type estimates alone.

(An earlier draft used a *Pythagorean* refinement of these two bounds, claiming $C_3^{(2)} < 1$; that refinement is withdrawn — the graded anticommutation does not produce operator-norm orthogonality — and replaced throughout by the rigorous triangle bound, giving $C_3^{(2)} \leq \sqrt{2}$; see Remark 4.5.)

That abandoned route's mechanism was unified: the same KO-dim-6 graded structure that defines the Connes–Marcolli composed Dirac split both its joint Lipschitz constant (R1) and its joint cross-Stein–Weiss panel bound (R2) into Leibniz terms controlled by the triangle bound, each of the five single-factor lemmas of [20] extending factor-by-factor. The live proof, by contrast, is the lifted-state assembly, which needs none of this (Lemma 4.12).

Why this is the *easy* case. The same-manifold tensor product treated here is the most accessible of three plausible tensor-product extensions of [20]. The cross-manifold case $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\mathrm{Bargmann}}$, in which the second factor is a holomorphic Hardy-sector spectral triple of the type used in the Bargmann–Segal lattice construction [19], is structurally obstructed at the framework level: the S^3 factor is a second-order Riemannian spinor Dirac, while the S^5 Bargmann factor is a first-order complex-analytic Euler operator on a Hardy sector and does not fit standard NCG Dirac axioms. There is no published prescription for tensoring a Riemannian spinor Dirac with a Hardy-sector first-order complex operator in the propinquity framework, and the obstruction is the four-layer Coulomb/HO asymmetry of [19] §V resurfacing at the spectral-triple level. A separate variant — $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\mathrm{Riem}}$ with both factors Riemannian — is mathematically clean (KO-arithmetic $3 + 5 = 8 \equiv 0 \pmod 8$) but requires propinquity convergence for the rank-2 $\mathcal{T}_{S^5}^{\mathrm{Riem}}$ as a separate prerequisite, which is not available in the literature. We address only the same-manifold case in the present paper, and discuss the obstructions to the cross-manifold extensions in §6.

Outline. Section 2 sets up the rescaled Camporesi–Higuchi spectral triple at focal length λ, the truncations, and the tensor-product Connes–Marcolli graded structure. Section 3 states the main theorem. Section 4 proves the tensor lemmas. Section 5 reports the numerical verification panel. Section 6 places the result in the Connes–van Suijlekom / Latrémolière / Farsi–Latrémolière / Hekkelman–McDonald lineage and discusses obstructions to the cross-manifold extensions. Section 7 lists open questions. Section 8 concludes.

2 The setting

Convergence convention and descope note. The object established here is van Suijlekom's *state-space Gromov–Hausdorff distance* (the convention of Paper 38 [20]), *not* the strictly stronger Latrémolière propinquity [12] (Adv. Math. **404** (2022), Paper No. 108393, preprint `arXiv:1811.10843`). The bound symbol $\Lambda(\cdot,\cdot)$ below denotes the state-space GH distance. The live route (the lifted-state assembly, Lemma 4.12) uses no dual reach, so reach_P does not enter it; the object is state-space GH rather than Latrémolière propinquity because the latter is not pursued here, and its dual-reach step would reintroduce the single-factor reach_P named gap (the invalid cb-norm dual-reach that descoped Papers 38/40). The abandoned (B,P)-pair assembly (Lemma 4.14) inherited that gap directly; its joint Lipschitz constant $C_3^{(2)} \le \sqrt{2}$ (Lemma 4.6) is that route's constant, not a rate constant of Theorem 3.1.

2.1 Single-factor recall: the rescaled S^3 Camporesi–Higuchi triple

For $\lambda > 0$, the *λ-scaled Camporesi–Higuchi spectral triple* on the unit-round $S^3 = \mathrm{SU}(2)$ is

$$\mathcal{T}^{\lambda}_{S^3} \;=\; \big(C^\infty(S^3),\; L^2(S^3,\Sigma),\; \lambda^{-1}D_{\mathrm{CH}}\big), \tag{4}$$

with D_{CH} as in [20] §2.1, eigenvalues $\sigma(\lambda^{-1}D_{\mathrm{CH}}) = \{\pm\lambda^{-1}(n+1/2) : n \in \mathbb{N}\}$ on each chirality sector. The truncated triple at cutoff $n_{\max}$ is $\mathcal{T}^{\lambda}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{n_{\max}}, \lambda^{-1}D_{n_{\max}})$, with the truncated operator system $\mathcal{O}_{n_{\max}} = P_{n_{\max}}C^\infty(S^3)P_{n_{\max}}$ and the truncated Dirac $D_{n_{\max}} = P_{n_{\max}}D_{\mathrm{CH}}P_{n_{\max}}$. We will write $\mathcal{T}^{\lambda}$ when the dependence on λ matters; the rescaling multiplies single-factor Connes distances and propinquity bounds by λ, but is otherwise transparent for our purposes.

2.2 The tensor-product spectral triple

Let $\lambda_a, \lambda_b > 0$. Define

$$\mathcal{T}^{\lambda_a}_{S^3} \otimes \mathcal{T}^{\lambda_b}_{S^3} \;:=\; \big(\mathcal{A}_{a,b},\; \mathcal{H}_{a,b},\; D_{a,b},\; \gamma_{a,b},\; J_{a,b}\big), \tag{5}$$

with

$$\mathcal{A}_{a,b} = C^\infty(S^3)\otimes C^\infty(S^3) = C^\infty(S^3\times S^3), \tag{6}$$
$$\mathcal{H}_{a,b} = L^2(S^3,\Sigma)\otimes L^2(S^3,\Sigma), \tag{7}$$
$$D_{a,b} = \lambda_a^{-1}D^{(a)}_{\mathrm{CH}}\otimes I_b + \gamma_a \otimes \lambda_b^{-1}D^{(b)}_{\mathrm{CH}}, \tag{8}$$
$$\gamma_{a,b} = \gamma_a \otimes \gamma_b, \tag{9}$$
$$J_{a,b} = J_a \otimes J_b \tag{10}$$

in the standard Connes–Marcolli graded tensor product convention [5, 16]. The crucial structural identity is

$$\big\{D^{(a)}_{\mathrm{CH}}\otimes I_b,\; \gamma_a\otimes D^{(b)}_{\mathrm{CH}}\big\} \;=\; \big(D^{(a)}_{\mathrm{CH}}\gamma_a + \gamma_a D^{(a)}_{\mathrm{CH}}\big)\otimes D^{(b)}_{\mathrm{CH}} \;=\; 0, \tag{11}$$

since $\{D^{(a)}_{\mathrm{CH}},\gamma_a\} = 0$ on the unit-round S^3 Dirac — an identity that holds only *vacuously*, γ_a being necessarily 0 [retracted 2026-09-04: p39-chirality-grading-on-s3] (see the note after (8)), so this structural identity carries no content as printed. This anticommutation of the two *Dirac* summands is what gives the clean two-term graded Leibniz rule (16) (it does *not*, however, make the two resulting *commutator* terms operator-norm orthogonal — see Remark 4.5); the joint commutator is then controlled by the triangle bound, which suffices to make the tensor-product propinquity convergent.

The truncated tensor-product triple at cutoffs $(n_{\max,a}, n_{\max,b})$ is

$$\mathcal{T}^{\lambda_a}_{n_{\max,a}} \otimes \mathcal{T}^{\lambda_b}_{n_{\max,b}} := \big(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}},\ \mathcal{H}_{n_{\max,a}} \otimes \mathcal{H}_{n_{\max,b}},\ P_{n_{\max,a}} \otimes P_{n_{\max,b}} \cdot D_{a,b} \cdot P_{n_{\max,a}} \otimes P_{n_{\max,b}}\big), \tag{12}$$

with the algebraic tensor-product operator system $\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}} \subset M_{\dim \mathcal{H}_{n_{\max,a}}}(\mathbb{C}) \otimes M_{\dim \mathcal{H}_{n_{\max,b}}}(\mathbb{C})$ defined as the linear span of Kronecker products $M_a \otimes M_b$ with $M_\bullet \in \mathcal{O}_{n_{\max,\bullet}}$.

3 Main theorem

Theorem 3.1 (Main theorem). ***[INTERNAL THEOREM]*** *The truncated tensor-product Camporesi–Higuchi spectral triples converge to the tensor product of the two continuum S^3 Camporesi–Higuchi spectral triples in van Suijlekom's state-space Gromov–Hausdorff distance (the strictly stronger Latrémolière propinquity is not claimed):*

$$\Lambda\big(\mathcal{T}^{\lambda_a}_{n_{\max,a}} \otimes \mathcal{T}^{\lambda_b}_{n_{\max,b}},\ \mathcal{T}^{\lambda_a}_{S^3} \otimes \mathcal{T}^{\lambda_b}_{S^3}\big) \ \le\ \gamma^{(ab)}_\lambda \ \le\ \lambda_a\, \gamma_{n_{\max,a}} + \lambda_b\, \gamma_{n_{\max,b}} \ \le\ 2\max\big(\lambda_a\, \gamma_{n_{\max,a}},\ \lambda_b\, \gamma_{n_{\max,b}}\big), \tag{13}$$

*where $\gamma^{(ab)}_\lambda = \iint K_{n_{\max,a}} K_{n_{\max,b}} \sqrt{\lambda_a^2\, d(e,g)^2 + \lambda_b^2\, d(e,h)^2}\, dg\, dh$ is the joint first moment and $\gamma_n = (4/\pi)\log n_{\max}/n_{\max} + O(1/n_{\max})$ (dual-Coxeter; $2/\pi$ on the unit S^3). **The leading constant is** 1: the tight bound is $\gamma^{(ab)}_\lambda$ itself, of which $\lambda_a\gamma_{n_{\max,a}} + \lambda_b\gamma_{n_{\max,b}}$ is the clean subadditive form and $2\max$ the looser corollary ($\gamma^{(ab)}$ is strictly below $\gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}$ by $\sqrt{x^2+y^2} \le x+y$, and strictly below $2\max$ at every finite cutoff). The right-hand side is $O(\log n_{\max}/n_{\max}) \to 0$ with universal asymptotic constant $4/\pi$ per factor.*

Metrization (the tier's scope). *As in Paper 38's* `thm:main_unconditional`*, both state spaces are metrized by the left-*translation *Lipschitz seminorm, which agrees with the truncated-Dirac commutator seminorm on the continuum (Lemma 4.11) and, unlike it, is non-degenerate at finite cutoff. The statement is unconditional* in that metrization *— Paper 38's own footing and the tier this theorem shares — not simpliciter.*

Route. *The bound is the lifted-state assembly of Lemma 4.12, which uses neither a Berezin map nor any partial inverse:* reach_P *is* unnecessary*, both almost-inverse defects being the same Fejér smoothing at the same moment $\gamma^{(ab)}$. The joint Lipschitz comparison constant $C_3^{(2)} \le \sqrt{2}$ of Lemma 4.6 belongs to the* abandoned *(B,P)-pair route (Lemma 4.14) and is not a rate constant of this theorem; its value is unchanged, only its role.*

Limit object. *$\mathcal{T}^{\lambda_a}_{S^3} \otimes \mathcal{T}^{\lambda_b}_{S^3}$ is the graded tensor product of the two odd (KO-3) Camporesi–Higuchi real spectral triples, with Dirac $D^{(a)}_{\mathrm{CH}} \otimes \sigma_1 \otimes 1 + 1 \otimes \sigma_2 \otimes D^{(b)}_{\mathrm{CH}}$, grading $1 \otimes \sigma_3 \otimes 1$ and real structure $J_{a,b} = J_a \otimes \sigma_1 K \otimes J_b$ at KO-dimension 6 (Lemma 4.13). The Clifford $\mathbb{C}^2$ is the factor carrying the product grading and real structure — the content of $3+3=6$; the displayed $D^{(a)}_{\mathrm{CH}} \otimes I + \gamma_a \otimes D^{(b)}_{\mathrm{CH}}$ is* not *realisable ($\gamma_a = 0$ on S^3, note after (8)), and the doubling is its correct replacement.*

History (2026-09-04/05). *Until 2026-09-04 this theorem was stated as established via the (B,P)-pair route (Lemma 4.14); that route's height leg was then refuted and its* reach_P *leg found to be a named gap, so the theorem was briefly re-tiered **[CONDITIONAL]**. The lifted-state route above (proved and independently reviewed 2026-09-05) discharges it back to **[INTERNAL THEOREM]** in the translation-seminorm metrization, with constant* 1 *rather than the conditional* 2*.*

Proof. The distance bound (13) is the lifted-state assembly of Lemma 4.12: its four facts (i)–(iv) give a correspondence of distortion $\le 2\gamma^{(ab)}$ between the two state spaces, hence $d_{\mathrm{GH}} \le \gamma^{(ab)}$, with $\gamma^{(ab)} \le \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}$ by Lemma 4.12(b). The substrate is Lemma 4.1 (operator system) and the joint kernel and rate are Lemma 4.3. The λ^{-1}-rescalings of the single-factor Diracs multiply each Monge–Kantorovich distance by λ, giving the $\lambda\gamma$ form. The identification of the

limit as the tensor product (standard odd⊗odd construction) is Lemma 4.13. The earlier (B, P)-pair route (Lemmas 4.6–4.14) is retained as the record of an abandoned approach: its height leg is refuted and its reach$_P$ leg a named gap, so it does not establish the bound; the lifted-state route replaces it and needs neither. □

4 The tensor lemmas

We work at unit focal length $\lambda_a = \lambda_b = 1$ throughout this section; the λ-rescaling is reinstated at the end of §4.5.

4.1 Lemma L1′-T: tensor-product operator system

Lemma 4.1 (L1′-T: tensor-product operator system substrate)**.** *Let $\mathcal{O}_{n_{\max,a}}$ and $\mathcal{O}_{n_{\max,b}}$ be the truncated operator systems of [20, §3.1] & [20, Lemma 3.1]. The algebraic tensor product*

$$\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}} \;\subset\; M_{\dim \mathcal{H}_{n_{\max,a}}}(\mathbb{C}) \otimes M_{\dim \mathcal{H}_{n_{\max,b}}}(\mathbb{C}),$$

defined as the linear span of Kronecker products $M_a \otimes M_b$ with $M_a \in \mathcal{O}_{n_{\max,a}}$, $M_b \in \mathcal{O}_{n_{\max,b}}$, satisfies:

(a) (Dimension factorization.) $\dim(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}) = \dim(\mathcal{O}_{n_{\max,a}}) \cdot \dim(\mathcal{O}_{n_{\max,b}})$.

(b) (Propagation bound.) $\mathrm{prop}(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}) \le 2$, *matching the single-factor Connes–van Suijlekom* $\mathrm{prop} = 2$ *value [6, §4.2].*

(c) (Cross-shell finite Connes distance.) Under the chirality-mixing offdiagonal extension $D^{\mathrm{offdiag}}_{a,b}$ defined factor-by-factor in the sense of [20, Lemma 3.1], every cross-shell pair of pure node-evaluation states has finite truncated joint Connes distance.

Proof. **(a)** The single-factor multiplier matrices $\{M_{NLM}\}$ are linearly independent in $M_{\dim \mathcal{H}_{n_{\max}},\bullet}(\mathbb{C})$ by construction ([20, §3.1]). The Kronecker product preserves linear independence: $\mathrm{vec}(M_a \otimes M_b) = \mathrm{vec}(M_a) \otimes \mathrm{vec}(M_b)$, and tensors of linearly independent vectors are linearly independent. Hence $\dim(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}) = \dim(\mathcal{O}_{n_{\max,a}}) \cdot \dim(\mathcal{O}_{n_{\max,b}})$.

(b) For any operator system $\mathcal{O}$ with $\mathrm{prop}(\mathcal{O}) = p$, $\mathcal{O}^p = M_{\dim \mathcal{H}}(\mathbb{C})$. For $p_a = p_b = 2$, we have $(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}})^2 \supset \mathcal{O}^2_{n_{\max,a}} \otimes \mathcal{O}^2_{n_{\max,b}} = M_{\dim \mathcal{H}_{n_{\max,a}}}(\mathbb{C}) \otimes M_{\dim \mathcal{H}_{n_{\max,b}}}(\mathbb{C}) = M_{\dim \mathcal{H}_{a,b}}(\mathbb{C})$, so $\mathrm{prop}(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}) \le 2$.

(c) The chirality-mixing offdiagonal extension $D^{\mathrm{offdiag}}_{a,b}$ inherits the cross-shell-finite property factor-by-factor from [20, Lemma 3.1]. For a state pair $(\varphi_a \otimes \varphi_b, \psi_a \otimes \psi_b)$ in distinct shells, at least one factor differs in shell, and the single-factor finite-distance property on that factor produces a non-trivial structural multiplier bounding the supremum in the SDP definition of the joint truncated Connes distance. □

Remark 4.2. As in [20, Remark 3.2], the offdiagonal extension is a metric-only device for finite-distance verification; the $D_{a,b} = D^{(a)}_{\mathrm{CH}} \otimes I + \gamma_a \otimes D^{(b)}_{\mathrm{CH}}$ printed here is what Lemmas 4.6–4.10 are proved for. It does *not* satisfy the KO-dim-6 axioms, and indeed cannot be realised: $\gamma_a = 0$ [retracted 2026-09-04: p39-chirality-grading-on-s3] (note after (8)). What those two lemmas establish for the repaired operator is open.

4.2 Lemma L2-T: joint central spectral Fejér kernel

Let $K^{(a)}_{n_{\max,a}}$ and $K^{(b)}_{n_{\max,b}}$ denote the single-factor central spectral Fejér kernels on SU(2) defined in [20, Definition 3.3]. Define the joint product kernel

$$K^{(a,b)}_{n_{\max,a},n_{\max,b}}(g_1, g_2) \;:=\; K^{(a)}_{n_{\max,a}}(g_1)\, K^{(b)}_{n_{\max,b}}(g_2) \qquad \text{on} \quad \mathrm{SU}(2) \times \mathrm{SU}(2). \tag{14}$$

Lemma 4.3 (L2-T: joint kernel construction and rate)**.** *The joint central kernel $K^{(a,b)}$ has the following properties:*

(a) (Positivity and unit mass.) $K^{(a,b)}(g_1, g_2) \geq 0$ *on* $\mathrm{SU}(2) \times \mathrm{SU}(2)$*, and* $\int (K^{(a,b)})\, dg_1\, dg_2 = 1$ *in the product Haar measure.*

(b) (Plancherel factorization.) The joint Plancherel symbol factorizes: $\widehat{K}^{(a,b)}(j_1, j_2) = \widehat{K}^{(a)}(j_1) \cdot \widehat{K}^{(b)}(j_2)$*, with each factor the closed-form* $\widehat{K}^{(\bullet)}(j) = (2j+1)/Z_{n_{\max,\bullet}}$ *for* $j \leq j_{\max,\bullet}$ *and zero otherwise.*

(c) (Cb-norm factorization.) $\|T_{K^{(a,b)}}\|_{\mathrm{cb}}^{\mathrm{cent}} = \|T_{K^{(a)}}\|_{\mathrm{cb}}^{\mathrm{cent}} \cdot \|T_{K^{(b)}}\|_{\mathrm{cb}}^{\mathrm{cent}} = 4/[(n_{\max,a}+1)(n_{\max,b}+1)]$.

(d) (Subadditive joint mass concentration.) Define $\gamma^{(a,b)}_{n_{\max,a},n_{\max,b}} := \int K^{(a,b)}(g_1, g_2)\, d_{S^3\times S^3}\big((e,e),(g_1,g_2)\big)\, dg_1\, d$ *Then*

$$\gamma^{(a,b)}_{n_{\max,a},n_{\max,b}} \;\leq\; \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}} \;\leq\; 2\,\max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}). \tag{15}$$

Proof. **(a)** The product of two non-negative unit-mass kernels is a non-negative unit-mass kernel under the product Haar measure. **(b)** The Plancherel decomposition of $L^2(G_1 \times G_2)$ is the product of the two single-factor Plancherel decompositions, and convolution by a central function on a product of compact groups acts diagonally on each isotypic component as the product of single-factor Fourier coefficients. **(c)** The Bożejko–Fendler equality $\|T_K\|_{\mathrm{cb}} = \|\widehat{K}\|_{\ell^\infty}$ for central convolutions on amenable compact groups [3, 17] extends to product kernels via the standard tensor-product cb-norm identity $\|A \otimes B\|_{\mathrm{cb}} = \|A\|_{\mathrm{cb}}\|B\|_{\mathrm{cb}}$ on the minimal tensor product [17, Ch. 8]. Substituting the single-factor result [20, Lemma 3.4(c)] gives the displayed value. **(d)** The product Riemannian metric on $S^3 \times S^3$ satisfies the triangle inequality $d_{S^3\times S^3}((e,e),(g_1,g_2)) \leq d_{S^3}(e,g_1) + d_{S^3}(e,g_2)$ at every point. Integrating against $K^{(a,b)}(g_1,g_2)\, dg_1\, dg_2$ and applying Fubini gives (15). □

Remark 4.4. Property (c) shows that the joint cb-norm is one order faster than the single-factor rate: $\|T_{K^{(a,b)}}\|_{\mathrm{cb}} = O(1/(n_{\max,a} n_{\max,b}))$ vs. single-factor $O(1/n_{\max})$. This is the structural reason joint convergence is well-behaved.

4.3 Lemma L3-T: joint Lipschitz comparison via the triangle bound on the graded Leibniz rule

This is the central new technical content of the paper. The Connes–Marcolli graded Dirac (8) on the KO-dim-6 graded module splits the joint commutator into two Leibniz terms (16); the operator-norm triangle bound on that sum is what controls the joint Lipschitz comparison constant.

Remark 4.5 (Withdrawn Pythagorean refinement)**.** An earlier version of this lemma claimed the *Pythagorean* identity $\|A+B\|_{\mathrm{op}}^2 = \|A\|^2 + \|B\|^2$ for the two Leibniz terms (citing the graded anticommutation $\{A,B\} = 0$), which would have given $C_3^{(2)} < 1$. That refinement is **false**: $\{A,B\} = 0$ does *not* imply operator-norm orthogonality (the elementary counterexample $A = \mathrm{diag}(1,0)$, $B = \mathrm{diag}(0,1)$ has $\{A,B\} = 0$ but $\|A+B\|^2 = 1 \neq 2 = \|A\|^2 + \|B\|^2$), and the specific factorisation that was invoked is invalid because the two Leibniz terms carry *different* second tensor factors ($M_g[D_{\mathrm{CH}}^{(b)}, M_g]$ vs. $[D_{\mathrm{CH}}^{(b)}, M_g]M_g$, which do not commute). A direct numerical test on the actual Camporesi–Higuchi harmonics (the supporting test below) finds the Pythagorean bound *maximally violated* ($\|A+B\|^2/(\|A\|^2+\|B\|^2) = 2$ exactly at a pair of top-shell raising harmonics, where the two terms add constructively) and the *triangle* bound tight ($\|A+B\| = \|A\| + \|B\|$ there). The constant is therefore $C_3^{(2)} \nearrow \sqrt{2}$, not < 1. This does not affect convergence: $\sqrt{2}$ is a finite constant, so the right-hand side of (13) still $\to 0$ as the cutoffs grow.

Lemma 4.6 (L3-T: joint Lipschitz comparison constant)**.** *For every simple-tensor multiplier $M_{f\otimes g} = M_f \otimes M_g$ with $f, g \in C^\infty(S^3)$, the joint Connes–Marcolli commutator satisfies the Leibniz identity*

$$\big[D_{a,b},\, M_f \otimes M_g\big] \;=\; [D_{\mathrm{CH}}^{(a)}, M_f] \otimes M_g + (\gamma_a M_f) \otimes [D_{\mathrm{CH}}^{(b)}, M_g], \tag{16}$$

and, by the operator-norm triangle inequality applied to the two Leibniz terms (which is tight *— see Remark 4.5),*

$$\big\|[D_{a,b}, M_f \otimes M_g]\big\|_{\mathrm{op}} \;\le\; \|[D_{\mathrm{CH}}^{(a)}, M_f]\|_{\mathrm{op}} \cdot \|M_g\|_{\mathrm{op}} \;+\; \|M_f\|_{\mathrm{op}} \cdot \|[D_{\mathrm{CH}}^{(b)}, M_g]\|_{\mathrm{op}}. \tag{17}$$

Combining (17) with the single-factor L3 estimate [20, Lemma 3.5] for each factor at level (N_a, N_b) in the Avery 3-Y harmonic basis gives the per-irrep ratio

$$\frac{\|[D_{a,b},\, M_{N_aL_aM_a} \otimes M_{N_bL_bM_b}]\|_{\mathrm{op}}^2}{\|Y^{(3)}_{N_aL_aM_a} \otimes Y^{(3)}_{N_bL_bM_b}\|_{\mathrm{Lip}}^2} \;\le\; \frac{\big((N_a-1)+(N_b-1)\big)^2}{N_a^2+N_b^2-2}. \tag{18}$$

Taking square roots and supping over the irrep grid gives the joint Lipschitz comparison constant

$$C_3^{(2)}(n_{\max,a}, n_{\max,b}) \;:=\; \sup_{2\le N_a\le n_{\max,a}+1\;\; 2\le N_b\le n_{\max,b}+1} \sqrt{\frac{\big((N_a-1)+(N_b-1)\big)^2}{N_a^2+N_b^2-2}} \;\le\; \sqrt{2}, \tag{19}$$

bounded by $\sqrt{2}$ at every finite $(n_{\max,a}, n_{\max,b})$ and increasing to $\sqrt{2}$ as the cutoffs grow. This constant prices the truncated-Dirac → classical-gradient conversion of the abandoned (B,P)-pair route (Lemma 4.14); it is not *a rate constant of Theorem 3.1, whose lifted-state route (via the translation seminorm, Lemma 4.11) never performs that conversion. The earlier Pythagorean claim $C_3^{(2)} < 1$ is withdrawn (Remark 4.5).*

Proof. The Leibniz identity (16) is a direct computation on Kronecker products:

$$[D_{a,b}, M_f \otimes M_g] = [D_{\mathrm{CH}}^{(a)} \otimes I_b, M_f \otimes M_g] + [\gamma_a \otimes D_{\mathrm{CH}}^{(b)}, M_f \otimes M_g],$$

with $[D_{\mathrm{CH}}^{(a)} \otimes I_b, M_f \otimes M_g] = [D_{\mathrm{CH}}^{(a)}, M_f] \otimes M_g$ (since I_b commutes with M_g), and $[\gamma_a \otimes D_{\mathrm{CH}}^{(b)}, M_f \otimes M_g] = \gamma_a M_f \otimes [D_{\mathrm{CH}}^{(b)}, M_g]$ (since $\gamma_a M_f \neq M_f \gamma_a$ in general but the chirality grading commutes with the scalar multiplier as a chirality-blind operator; in components, the chirality flip is absorbed into the γ_a factor on the left).

The bound (17) is the operator-norm triangle inequality $\|A+B\|_{\mathrm{op}} \le \|A\| + \|B\|$ applied to the two Leibniz terms $A = [D_{\mathrm{CH}}^{(a)}, M_f] \otimes M_g$ and $B = (\gamma_a M_f) \otimes [D_{\mathrm{CH}}^{(b)}, M_g]$, together with the multiplicativity of the operator norm on simple tensors ($\|X \otimes Y\| = \|X\|\,\|Y\|$, and $\|\gamma_a M_f\| = \|M_f\|$ since γ_a is unitary). This bound is *tight*: on a pair of top-shell raising harmonics the two terms align so that $\|A+B\| = \|A\| + \|B\|$ exactly. There $\|A\| = \|B\|$, so $\|A+B\|^2 = (2\|A\|)^2 = 4\|A\|^2$ while the Pythagorean value is $\|A\|^2 + \|B\|^2 = 2\|A\|^2$ — a factor 2 violation of the would-be identity (see Remark 4.5 for why the graded anticommutation does *not* give operator-norm orthogonality). The numerical verification on the actual Camporesi–Higuchi harmonics — the joint commutator built from the real D_{CH} matrices and the graded multipliers at $(n_{\max,a}, n_{\max,b}) = (2,2), (2,3)$, with the operator norm computed by singular-value decomposition — finds, over every harmonic pair, the triangle ratio $\|A+B\|/(\|A\|+\|B\|)$ attaining its maximum 1 and the Pythagorean ratio $\|A+B\|^2/(\|A\|^2+\|B\|^2)$ attaining its maximum 2, confirming both the tightness of (17) and the failure of the Pythagorean form (backing test `tests/test_paper39_triangle_tight.py`).

For the per-irrep ratio (18), the single-factor L3 estimate [20, Eq. (3.18)] gives $\|[D_{\mathrm{CH}}^{(\bullet)}, M_{N_\bullet L_\bullet M_\bullet}]\|_{\mathrm{op}} \le (N_\bullet - 1)$ on the unit-normalised Avery harmonic $Y^{(3)}_{N_\bullet L_\bullet M_\bullet}$. The joint Lipschitz norm satisfies

$$\|Y^{(3)}_{N_aL_aM_a} \otimes Y^{(3)}_{N_bL_bM_b}\|_{\mathrm{Lip}}^2 \;=\; (N_a^2-1)\cdot 1 + 1\cdot(N_b^2-1) \;=\; N_a^2+N_b^2-2, \tag{20}$$

where we have used $\|Y^{(3)}_{N_aL_aM_a}\|^2_{\mathrm{Lip}} = N_a^2 - 1$ (the round-S^3 Laplace–Beltrami eigenvalue at level N_a) and $\|Y^{(3)}_{N_aL_aM_a}\|_\infty \le 1$ on the unit-normalised panel. Combining (17) and (20) gives (18).

The supremum over the irrep rectangle $[2, n_{\max,a}+1] \times [2, n_{\max,b}+1]$ in (19) is finite and bounded by $\sqrt{2}$: since $\big((N_a-1)+(N_b-1)\big)^2 \le 2\big((N_a-1)^2+(N_b-1)^2\big) \le 2(N_a^2+N_b^2-2)$, the per-irrep ratio never exceeds 2. The supremum is computed by scanning the (finite) irrep grid; it equals 1 at $(2,2)$, exceeds 1 for every larger cutoff, and increases to $\sqrt{2} \approx 1.414$ as $n_{\max,a}, n_{\max,b} \to \infty$ (diagonal values 1.000 at $(2,2)$, 1.095 at $(3,3)$, 1.155 at $(4,4)$; see Remark 4.7). □

Remark 4.7 (Diagonal closed form)**.** On the diagonal $n_{\max,a} = n_{\max,b} = n$, the supremum in (19) is attained at the corner $(n+1, n+1)$ and reads

$$C_3^{(2)}(n,n) \;=\; \sqrt{\frac{(2n)^2}{2(n+1)^2-2}} \;=\; \sqrt{\frac{2n}{n+2}}.$$

Numerical values: $C_3^{(2)}(2,2) = 1$, $C_3^{(2)}(3,3) = \sqrt{6/5} \approx 1.095$, $C_3^{(2)}(10,10) \approx 1.291$, $C_3^{(2)}(100,100) \approx 1.400$, with $C_3^{(2)}(n,n) \nearrow \sqrt{2}$. The constant exceeds 1 for every $n \ge 3$; the convergence in (13) is nonetheless driven entirely by the factor $\max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}) \to 0$, which the bounded constant $C_3^{(2)} \le \sqrt{2}$ does not disturb.

Remark 4.8 (Asymmetric supremum)**.** For asymmetric cutoffs $n_{\max,a} \ne n_{\max,b}$, the supremum is again bounded by $\sqrt{2}$ and is found by scanning the irrep grid. At $(n_{\max,a}, n_{\max,b}) = (10,4)$ for example, the maximum over $[2,11] \times [2,5]$ is ≈ 1.179 at the *interior* point $(8,5)$ ($\sqrt{121/87}$) — the supremum need not occur at the grid corner ($(11,5)$ gives only $\sqrt{196/144} \approx 1.167$). In all cases $C_3^{(2)} \le \sqrt{2}$, which is what the convergence theorem requires.

4.4 Lemma L4-T: joint Berezin reconstruction

Definition 4.9 (Joint Berezin reconstruction)**.** For $n_{\max,a}, n_{\max,b} \ge 1$ and $f, g \in C^\infty(S^3)$, define

$$B_{n_{\max,a},n_{\max,b}}(f \otimes g) \;:=\; B_{n_{\max,a}}(f) \otimes B_{n_{\max,b}}(g), \tag{21}$$

and extend to general $h \in C^\infty(S^3 \times S^3)$ by linear extension on the dense subspace of finite linear combinations of simple tensors.

Lemma 4.10 (L4-T: joint Berezin properties)**.** *The map $B_{n_{\max,a},n_{\max,b}} : C^\infty(S^3 \times S^3) \to \mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}$ has the following four properties on simple tensors (and extends to positive cones / by density):*

(a) (Positivity.) If $f, g \ge 0$ on S^3, then $B_{n_{\max,a},n_{\max,b}}(f \otimes g)$ is positive semi-definite.

(b) (Contractivity.) $\|B_{n_{\max,a},n_{\max,b}}(f \otimes g)\|_{\mathrm{op}} \le \|f\|_\infty \cdot \|g\|_\infty = \|f \otimes g\|_\infty$.

(c) (Approximate identity.)

$$\|B_{n_{\max,a},n_{\max,b}}(f\otimes g) - (P_{n_{\max,a}}\otimes P_{n_{\max,b}}) M_{f\otimes g}(P_{n_{\max,a}}\otimes P_{n_{\max,b}})\|_{\mathrm{op}} \;\le\; \gamma_{n_{\max,a}}\|g\|_\infty\|\nabla f\|_\infty + \|f\|_\infty \gamma_{n_{\max,b}}\|$$

(d) (Compatibility with L3-T.) $\|[D_{a,b}, B_{n_{\max,a},n_{\max,b}}(f \otimes g)]\|_{\mathrm{op}} \le C_3^{(2)} \|f \otimes g\|_{\mathrm{Lip}}$, with $C_3^{(2)} \le \sqrt{2}$ given by (19).

Proof. Each property factorizes from the single-factor property [20, Lemma 3.6]. **(a)** The Kronecker product of two positive semi-definite operators is positive semi-definite. **(b)** $\|B_{n_{\max,a}}(f) \otimes$

$B_{n_{\max,b}}(g)\|_{\mathrm{op}} = \|B_{n_{\max,a}}(f)\|_{\mathrm{op}} \cdot \|B_{n_{\max,b}}(g)\|_{\mathrm{op}} \le \|f\|_\infty \cdot \|g\|_\infty$ by single-factor L4(b) on each side. **(c)** Use the telescoping identity

$$B_a(f)\otimes B_b(g) - P_a M_f P_a \otimes P_b M_g P_b = \big(B_a(f) - P_a M_f P_a\big)\otimes B_b(g) + P_a M_f P_a \otimes \big(B_b(g) - P_b M_g P_b\big),$$

take operator norms, and apply single-factor L4(c) on each summand. **(d)** Apply Lemma 4.6 to the simple tensor $B_a(f)\otimes B_b(g) \in \mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}$, using single-factor L4(d) for the per-factor commutator bound and the triangle bound (17) for the operator-norm reduction. □

4.5 Lemma L5-T: joint state-space-GH assembly

4.6 Lemma continuum-Lip-T: translation seminorm equals classical Lipschitz

Lemma 4.11 (continuum-Lip-T). ***[INTERNAL THEOREM]*** *For $F \in C^\infty(S^3\times S^3)$, $L_{\mathrm{prod}}(F) = \mathrm{Lip}_{d_{\mathrm{prod}}}(F) = \|\nabla_{\mathrm{prod}} F\|_\infty$, where L_{prod} is the left-translation seminorm on the product group $\mathbb{G} = \mathrm{SU}(2)\times\mathrm{SU}(2)$ and d_{prod} the product geodesic metric.*

Proof. $\|\lambda_u F - F\|_\infty \le \mathrm{Lip}(F)\, d_{\mathrm{prod}}(e,u)$ by bi-invariance, so $L_{\mathrm{prod}} \le \mathrm{Lip}$; conversely left translations act transitively and realise every geodesic segment, so the supremum recovers Lip. This is Paper 38's `lem:continuum_lip` [20] with $\mathbb{G}$ in place of SU(2). With the doubled Dirac (Lemma 4.13) the third equality $\|\nabla_{\mathrm{prod}} F\|_\infty = \|[D_{a,b}, M_F]\|_{\mathrm{op}}$ also holds exactly: this is where a Lipschitz-comparison constant $C_3^{(2)}$ would enter, and on the continuum symbol it is 1. (The truncated-cutoff $C_3^{(2)} \le \sqrt{2}$ of Lemma 4.6 prices the *truncated*-Dirac → classical gradient conversion of the abandoned (B,P) route, which this route does not perform.) □

4.7 Lemma lifted-state-T: the lifted-state route

Let $J_\bullet = (n_\bullet - 1)/2$, $h_\bullet = \sum_{j\le J_\bullet} \sqrt{2j+1}\,\chi_j$, $\chi_\bullet \in \mathbb{C}^2$ unit spinors, $\xi_\bullet = (h_\bullet \otimes \chi_\bullet)/\sqrt{Z_{n_\bullet}}$, $\eta \in \mathbb{C}^2$ unit, and set

$$\xi_{a,b} := \xi_a \otimes \eta \otimes \xi_b, \qquad \upsilon_{a,b}(T)(u) := \langle U_u\, \xi_{a,b},\, T\, U_u\, \xi_{a,b}\rangle, \quad u = (g,h) \in \mathbb{G}. \tag{22}$$

Lemma 4.12 (lifted-state-T). *$\upsilon_{a,b}$ and the compression $S_{a,b}(F) = P_{a,b} M_F P_{a,b}$ form an approximation pair for which:* (a) *$\xi_{a,b} \in \mathrm{ran}\, P_{a,b}$;* (b) *the induced measure is the product Fejér measure, with $\langle \xi_{a,b}, S_{a,b}(F)\,\xi_{a,b}\rangle = \iint F\, K_{n_{\max,a}} K_{n_{\max,b}}$ and first moment $\gamma^{(ab)} \le \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}$;* (c) *$L_{\mathrm{prod}}(\upsilon_{a,b}(T)) \le L_{a,b}(T)$ (tautological; no $C_3^{(2)}$, no graded Leibniz). The four tunnel facts*

$$\begin{aligned} &\textit{(i)}\ L_{a,b}(S_{a,b}F) \le L_{\mathrm{prod}}(F), && \textit{(ii)}\ L_{\mathrm{prod}}(\upsilon_{a,b}T) \le L_{a,b}(T),\\ &\textit{(iii)}\ \|\upsilon_{a,b}(S_{a,b}F) - F\|_\infty \le \gamma^{(ab)} L_{\mathrm{prod}}(F), && \textit{(iv)}\ \|S_{a,b}(\upsilon_{a,b}T) - T\|_{\mathrm{op}} \le \gamma^{(ab)} L_{a,b}(T) \end{aligned}$$

*hold, and **[INTERNAL THEOREM]** the correspondence $R = \{(\omega, S_{a,b}^*\omega)\} \cup \{(\upsilon_{a,b}^*\phi, \phi)\}$ has distortion $\le 2\gamma^{(ab)}$, so*

$$\Lambda\big(\mathcal{T}_{n_{\max,a}} \otimes \mathcal{T}_{n_{\max,b}},\ \mathcal{T}_{S^3}\otimes\mathcal{T}_{S^3}\big) \;=\; d_{\mathrm{GH}} \;\le\; \gamma^{(ab)} \;\le\; \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}. \tag{23}$$

reach_P is unnecessary: both almost-inverse defects (iii),(iv) are the same Fejér smoothing at the same moment $\gamma^{(ab)}$.

Proof. *(a)* $\xi_a \in \mathrm{ran}\, P_{n_{\max,a}}$, $\xi_b \in \mathrm{ran}\, P_{n_{\max,b}}$ (Paper 38 `lem:lifted_state`(a)), and η is untouched by 1_2; the doubling $\mathbb{C}^2$ carries the Clifford structure of the product Dirac, not the algebra or the group, so it is inert for the state. *(b)* $P_{a,b}\xi_{a,b} = \xi_{a,b}$ gives the product-measure identity; $\gamma^{(ab)} = \iint K_{n_{\max,a}} K_{n_{\max,b}} \sqrt{d(e,g)^2 + d(e,h)^2} \le \iint KK\,(d(e,g) + d(e,h)) = \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}$;

both sides scale together under the metric normalisation, so the statement is normalisation-independent. *(c)* With $\upsilon_{a,b}(T)(u) = \langle \xi_{a,b}, \rho_{u^{-1}}(T)\xi_{a,b}\rangle$ and $\rho_{u^{-1}} - \rho_{u'^{-1}} = \rho_{u'^{-1}} \circ (\rho_{u'u^{-1}} - \mathrm{id})$, $|\upsilon_{a,b}(T)(u) - \upsilon_{a,b}(T)(u')| \le \|\rho_q(T) - T\|_{\mathrm{op}} \le L_{a,b}(T)\, d_{\mathrm{prod}}(u,u')$ by bi-invariance; the graded Leibniz rule never appears because the seminorm is a translation seminorm. *(i)* $\rho_u(S_{a,b}F) - S_{a,b}F = S_{a,b}(\lambda_u F - F)$ and compression contracts operator norms. *(ii)* is (c). *(iii)* is the good-kernel estimate $|K * F(x) - F(x)| \le \int K(u)|F(u^{-1}x) - F(x)| \le \mathrm{Lip}(F)\int K\, d_{\mathrm{prod}} = \gamma^{(ab)}\mathrm{Lip}(F)$. *(iv)* on $T = P_{a,b}M_F P_{a,b}$, $S_{a,b}(\upsilon_{a,b}T) = P_{a,b}M_{K*F}P_{a,b} = \int \rho_u(T) K_{n_{\max,a}} K_{n_{\max,b}}(u)\, du =: \Phi_{a,b}(T)$, and $\|\Phi_{a,b}(T) - T\|_{\mathrm{op}} \le \int \|\rho_u(T) - T\|_{\mathrm{op}} KK \le L_{a,b}(T)\gamma^{(ab)}$. The four-pair distortion computation (both-first/both-second/two mixed) gives $\mathrm{dis}(R) \le 2\gamma^{(ab)}$, hence $d_{\mathrm{GH}} \le \frac{1}{2}\mathrm{dis}(R) = \gamma^{(ab)}$. This is Paper 38's `thm:main_unconditional` proof, lifted factor-wise to $\mathbb{G}$; facts (i),(ii) hold for *all* operators, so the bound covers the full (entangled-inclusive) joint state space. □

4.8 Lemma real-structure-T: the KO-6 tensor triple

Lemma 4.13 (real-structure-T). *The Clifford-doubled product* $\big(\mathcal{A}_a \otimes 1_2 \otimes \mathcal{A}_b,\ \mathcal{H}_a \otimes \mathbb{C}^2 \otimes \mathcal{H}_b,\ D^{(a)}_{\mathrm{CH}} \otimes \sigma_1 \otimes 1 + 1 \otimes \sigma_2 \otimes D^{(b)}_{\mathrm{CH}},\ 1 \otimes \sigma_3 \otimes 1\big)$ *carries the real structure*

$$J_{a,b} \;=\; J_a \otimes \sigma_1 K \otimes J_b \tag{24}$$

[INTERNAL THEOREM] *(K = complex conjugation on $\mathbb{C}^2$, $J_\bullet$ the single-factor Camporesi–Higuchi charge conjugation), satisfying the KO-dimension-6 signs*

$$J^2_{a,b} = +1, \qquad J_{a,b}\, D_{a,b} = +\, D_{a,b}\, J_{a,b}, \qquad J_{a,b}\, \gamma_{a,b} = -\,\gamma_{a,b}\, J_{a,b}, \tag{25}$$

together with the order-zero and order-one conditions in the continuum limit. The middle factor $\sigma_1 K$ is forced: of the four antiunitaries WK, $W \in \{1, \sigma_1, \sigma_2, \sigma_3\}$, only σ_1 yields (25) (σ_2 gives $J^2 = -1$; 1 and σ_3 fail $J_{a,b}D_{a,b} = D_{a,b}J_{a,b}$ on the second Dirac term). This is the graded tensor product of the two odd (KO-3) Camporesi–Higuchi real spectral triples, so the limit is the *tensor product, not merely* a *tensor triple.*

Proof. On the Clifford $\mathbb{C}^2$ factor: $J^2_{a,b} = J^2_a \otimes (\sigma_1 K)^2 \otimes J^2_b = (-1)(\sigma_1\bar{\sigma}_1)(-1) = +\sigma_1^2 = +1$. For $J_{a,b}D_{a,b}$, on $D^{(a)}_{\mathrm{CH}} \otimes \sigma_1 \otimes 1$ the middle factor gives $\sigma_1 K \cdot \sigma_1 = \sigma_1\sigma_1 K$ vs $\sigma_1 \cdot \sigma_1 K$ (commute, +); on $1 \otimes \sigma_2 \otimes D^{(b)}_{\mathrm{CH}}$, $\sigma_1 K \cdot \sigma_2 = \sigma_1\bar{\sigma}_2 K = -\sigma_1\sigma_2 K$ while $\sigma_2 \cdot \sigma_1 K = \sigma_2\sigma_1 K = -\sigma_1\sigma_2 K$ (equal, +), using the KO-3 single-factor $J_\bullet D^{(\bullet)}_{\mathrm{CH}} = +D^{(\bullet)}_{\mathrm{CH}} J_\bullet$; hence $J_{a,b}D_{a,b} = +D_{a,b}J_{a,b}$. $J_{a,b}\gamma_{a,b}$: $\sigma_1 K \cdot \sigma_3 = \sigma_1\sigma_3 K = -\sigma_3\sigma_1 K = -\sigma_3 \cdot \sigma_1 K$, so $J_{a,b}\gamma_{a,b} = -\gamma_{a,b}J_{a,b}$. The three signs $(+,+,-)$ are the KO-6 row (van Suijlekom, and the $3+3=6$ additivity of the KO dimension). *Order-zero and order-one.* These reduce factor-by-factor: for $a = M_f \otimes 1_2 \otimes M_g$ and $b = M_{f'} \otimes 1_2 \otimes M_{g'}$ in $\mathcal{A}_{a,b}$, the middle 1_2 conjugates to 1_2 under $J_{a,b}$, so $[a, J_{a,b}\, b\, J^{-1}_{a,b}] = 0$ holds iff each leg's order-zero holds, and $[[D_{a,b}, a], J_{a,b}\, b\, J^{-1}_{a,b}] = 0$ reduces on each leg to the factor's order-zero and order-one (proved). In the continuum the limit algebra $C^\infty(S^3) \otimes 1_2 \otimes C^\infty(S^3)$ is *commutative*, for which both conditions are automatic (JbJ^{-1} is right multiplication, which commutes with left multiplication and with $[D, a]$; Paper 32 §VIII, the standard commutative fact) — so the continuum tensor triple satisfies them. **At finite cutoff they do not hold, and the residual *grows* with the cutoff** (measured max residual $5.5 \times 10^{-2} \to 7.9 \times 10^{-2}$ for order-zero and $1.0 \times 10^{-1} \to 2.0 \times 10^{-1}$ for order-one from $n_{\max} = 2$ to 3, and continuing to grow at higher cutoff): the truncated operator system is not an algebra, the same phenomenon Paper 32 §VIII (order zero and order one) records for the single factor. (An earlier version pointed to Paper 38 `rem:dirac_degeneracy`, which concerns the doubled model's commutator-seminorm kernel, not the order conditions; repointed 2026-09-28.) The product introduces *no cross-factor obstruction* — every cross-factor pair satisfies both conditions to machine precision, so the finite-cutoff defect is exactly the inherited single-factor one. The order conditions are *not* required for the distance bound of Theorem 3.1 (which uses only the state-space assembly); they bear only on

the identification of the *limit* as a real spectral triple, and there the commutative-limit argument settles them. Full sign table, the factor-reduction, the cross-factor check and the growing single-factor residual are verified in the backing test `tests/test_paper39_real_structure.py`. □

Lemma 4.14 (L5-T: joint state-space-GH bound from the joint approximation pair, conditional)**.** *The joint approximation pair* $(B_{n_{\max,a},n_{\max,b}}, P_{n_{\max,a}} \otimes P_{n_{\max,b}})$ *with* $B_{n_{\max,a},n_{\max,b}}$ *from Definition 4.9 and* $P_{n_{\max,a}} \otimes P_{n_{\max,b}}$ *the joint spectral compression is an approximation pair between* $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ *and* $\mathcal{T}_{n_{\max,a}}^{\lambda_a} \otimes \mathcal{T}_{n_{\max,b}}^{\lambda_b}$*, and yields,* conditionally on the reach_P named gap *(see the proof), the bound*

$$\Lambda\big(\mathcal{T}_{n_{\max,a}}^{\lambda_a} \otimes \mathcal{T}_{n_{\max,b}}^{\lambda_b},\ \mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}\big) \ \le\ C_3^{(2)} \cdot 2 \cdot \max\Big(\lambda_a\, \gamma_{n_{\max,a}},\ \lambda_b\, \gamma_{n_{\max,b}}\Big). \tag{26}$$

Proof. We work at unit focal length $\lambda_a = \lambda_b = 1$ and reinstate the rescaling at the end. The four constituents of a Latrémolière tunnel for metric spectral triples [12] on the joint pair are:

(i) $\mathrm{reach}_{B_{n_{\max,a},n_{\max,b}}}$;

(ii) $\mathrm{reach}_{P_{n_{\max,a}} \otimes P_{n_{\max,b}}}$;

(iii) $\mathrm{height}_{B_{n_{\max,a},n_{\max,b}}}$, the Lipschitz-distortion height (Lipschitz-seminorm-comparison form [20, Remark 3.8]);

(iv) $\mathrm{height}_{P_{n_{\max,a}} \otimes P_{n_{\max,b}}}$.

Only (i) and (ii) survive below; (ii) is a named gap, so the pair yields the bound only conditionally on it. (iii) and (iv) are listed because they are what a Latrémolière tunnel would additionally require, and both are shown below to be uncontrolled — see (29).

Reach. By Lemma 4.10(c) on the unit-Lipschitz panel,

$$\mathrm{reach}_{B_{n_{\max,a},n_{\max,b}}} \ \le\ \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}} \ \le\ 2\max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}).$$

The dual $\mathrm{reach}_{P_{n_{\max,a}} \otimes P_{n_{\max,b}}}$ inherits the same bound by symmetry of the joint convolution — and with it the single-factor reach_P *named gap* (the invalid cb-norm dual-reach that descoped Papers 38/40), so the bound below is at the state-space-GH level, not Latrémolière propinquity.

Height. By the Lipschitz-distortion form [12, §4] [20, Remark 3.8],

$$\mathrm{height}_{B_{n_{\max,a},n_{\max,b}}} \ =\ \sup_{\|h\|_{\mathrm{Lip}} \le 1} \Big|\, \|h\|_{\mathrm{Lip}} - \|B_{n_{\max,a},n_{\max,b}}(h)\|_{\mathrm{Lip}}^{\mathcal{O}} \Big|.$$

Taking the supremum on simple tensors $h = f \otimes g$ with $L_f := \|\nabla f\|_\infty$, $M_f := \|f\|_\infty$ (and similarly for g) and using $\|f \otimes g\|_{\mathrm{Lip}}^2 \le L_f^2 M_g^2 + M_f^2 L_g^2$ (the pointwise continuum-gradient identity on the product Riemannian metric — this Pythagorean form is for the *classical* gradient and is exact) together with the triangle bound (17) applied to the joint commutator, single-factor L4(d) on each factor, and the elementary identity $|x - y| = (x^2 - y^2)/(x + y)$ on $\|h\|_{\mathrm{Lip}}^2 - \|B(h)\|_{\mathrm{Lip}}^2$ gives the explicit bound

$$\mathrm{panel}_{B_{n_{\max,a},n_{\max,b}}}(f \otimes g) \ \le\ \frac{2L_f^2 M_g^2 \gamma_{n_{\max,a}} + 2L_g^2 M_f^2 \gamma_{n_{\max,b}}}{\sqrt{L_f^2 M_g^2 + M_f^2 L_g^2}}. \tag{27}$$

On the unit-norm panel ($L_f = L_g = M_f = M_g = 1$), this collapses to

$$\mathrm{panel}_{B_{n_{\max,a},n_{\max,b}}} \ \le\ \sqrt{2}\,(\gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}) \ \le\ 2\sqrt{2}\,\max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}). \tag{28}$$

Withdrawn 2026-09-04 [retracted 2026-09-04: l5-height-bound-achieved]. Until that date the two displays above were labelled height_B and were followed by $\mathrm{height}_{P_{n_{\max,a}} \otimes P_{n_{\max,b}}} = 0$,

and both were assembled into the rate. They are *panel* quantities — suprema over unit-norm simple tensors — and not the Lipschitz-distortion height, which is a supremum over the whole unit-Lipschitz ball. The witness Paper 38 [20] gives for the rank-one case transports verbatim: $B_{n_{\max,a},n_{\max,b}} = B_{n_{\max,a}} \otimes B_{n_{\max,b}}$ is a finite-band reconstruction, so taking $h = f \otimes \mathbf{1}$ with f spectrally supported above the multiplier envelope $N \le 2n_{\max,a} - 1$ of $B_{n_{\max,a}}$ (Paper 38, Definition `def:berezin`) and $\|h\|_{\mathrm{Lip}} = 1$ gives $B_{n_{\max,a},n_{\max,b}}(h) = 0$ and hence

$$\mathrm{height}_{B_{n_{\max,a},n_{\max,b}}} \equiv 1 \qquad \text{at every } (n_{\max,a}, n_{\max,b}), \tag{29}$$

with $\mathrm{height}_{P_{n_{\max,a}} \otimes P_{n_{\max,b}}} \neq 0$ by the same witness. The cross-Stein–Weiss computation (27) is unaffected and is retained as the panel statement it is; what is withdrawn is its use as a height constituent.

Assembly. With the height leg withdrawn, what remains of the pair is $\mathrm{reach}_B \le \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}$ (on the unit-Lipschitz panel, Lemma 4.10(c)) and reach_P, a named gap. (An earlier version of this paragraph called this "the footing on which Papers 38 and 40 stand"; it is not: Paper 38's surviving route is its lifted-state construction, which uses neither B nor a dual reach; this paper's tensor analogue of it is Lemma 4.12, the live proof of Theorem 3.1, which supersedes the conditional (B, P) assembly recorded in this lemma.) Conditionally on the reach_P gap, with the $C_3^{(2)}$ Lipschitz comparison constant of Lemma 4.6 (which controls the conversion between $\|\cdot\|_{\mathrm{Lip}}^{\mathcal{O}}$ in the truncated triple and the classical $\|\nabla\cdot\|_\infty$ on $S^3 \times S^3$), the reach paragraph above gives

$$\begin{aligned} \Lambda \;\le\; \max\bigl(\mathrm{reach}_B,\, \mathrm{reach}_P\bigr) \;&\le\; C_3^{(2)} \cdot (\gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}) \\ &\le\; C_3^{(2)} \cdot 2 \cdot \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}). \end{aligned}$$

The constant improves from the withdrawn $1 + 2\sqrt{2} \approx 3.828$ [retracted 2026-09-04: p39-tensor-assembly-constant] to 2, the refuted leg having been the larger of the two; it also corrects an arithmetic slip in the withdrawn assembly, which bounded reach by $\max(\gamma)$ where this paragraph derives $\gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}$. The λ^{-1}-rescalings of (8) multiply each Monge–Kantorovich distance by λ (§2.1), producing the $\max(\lambda\gamma)$-form of (26). □

Remark 4.15 (Identification of the cross-Stein–Weiss term)**.** The first-pass internal proof of this lemma sketched the cross-term contribution as $\varepsilon_{\mathrm{cross}} = O(\gamma_{n_{\max,a}} \gamma_{n_{\max,b}})$, suggesting a quadratic decay rate. The closed-form analysis above shows the actual rate is linear: $\varepsilon_{\mathrm{cross}} \le (2\sqrt{2} - 1) \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}) \approx 1.83 \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}})$ on the unit-norm panel. Since the height leg is withdrawn (2026-09-04), $\varepsilon_{\mathrm{cross}}$ no longer enters the rate at all; it survives as a unit-norm panel quantity, and the remark above is retained only for the record of what the first pass got wrong about its order.

5 Numerical verification

We verify the joint state-space-GH bound at the panel $(n_{\max,a}, n_{\max,b}) \in \{(2,2), (2,3), (3,3), (3,4), (4,4)\}$.

Single-factor input. The single-factor moments $\gamma_{n_{\max}}$ inherit from [20, Lemma 3.4]:

$$\gamma_2 = \pi - \tfrac{64\sqrt{2}}{27\pi} \approx 2.0746, \quad \gamma_3 \approx 1.6101, \quad \gamma_4 \approx 1.3223.$$

Joint cb-norm. By Lemma 4.3(c), the joint cb-norms are exact rationals: $\|T^{(2,2)}\|_{\mathrm{cb}} = 4/9$, $\|T^{(2,3)}\|_{\mathrm{cb}} = 1/3$, $\|T^{(3,3)}\|_{\mathrm{cb}} = 1/4$, $\|T^{(3,4)}\|_{\mathrm{cb}} = 1/5$, $\|T^{(4,4)}\|_{\mathrm{cb}} = 4/25$.

Joint Lipschitz comparison constant. By Lemma 4.6 (tight triangle bound), the diagonal sequence is

$$C_3^{(2)}(n,n) = \sqrt{2n/(n+2)} \nearrow \sqrt{2},$$

giving $C_3^{(2)}(2,2) = 1$, $C_3^{(2)}(3,3) = \sqrt{6/5} \approx 1.095$, $C_3^{(2)}(4,4) = \sqrt{8/6} \approx 1.155$.

Panel values for the abandoned (B,P)-pair route (unit focal length). The columns below tabulate the *abandoned* (B,P)-pair bound (Lemma 4.14), retained for the record; they are *not* Theorem 3.1's bound, which is the tighter $\gamma^{(ab)} \le \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}$ of constant 1, which is strictly below the Λ^{full} values here (those carry the extra $C_3^{(2)} \ge 1$ factor of the abandoned route). We report two presentations of the bound at the panel: the *factorised-panel* bound $\Lambda^{\text{fact}} = C_3^{(2),\text{fact}} \cdot \max(\gamma_a, \gamma_b)$ with $C_3^{(2),\text{fact}} = 1$ on the unit-Lipschitz simple-tensor panel (this is the variant computed by `compute_tensor_propinquity_bound` as `propinquity_bound`), and the *full-operator-system* bound (26) obtained by inserting the triangle-bound $C_3^{(2)} \le \sqrt{2}$ of Lemma 4.6 together with the factor 2 reach contribution of Lemma 4.14.

$(n_{\max,a}, n_{\max,b})$	$\gamma_{n_{\max,a}}$	$\gamma_{n_{\max,b}}$	$\|T^{(a,b)}\|_{\text{cb}}$	Λ^{fact}	$C_3^{(2)}$	Λ^{full}
$(2,2)$	2.0746	2.0746	4/9	2.0746	1.000	4.149
$(2,3)$	2.0746	1.6101	1/3	2.0746	1.043	4.326
$(3,3)$	1.6101	1.6101	1/4	1.6101	1.095	3.528
$(3,4)$	1.6101	1.3223	1/5	1.6101	1.121	3.610
$(4,4)$	1.3223	1.3223	4/25	1.3223	1.155	3.054

The factorised-panel column Λ^{fact} uses $\Lambda^{\text{fact}} \le \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}})$, the value tracked in the supporting computational module as the factorised-panel propinquity bound. The full-operator-system column Λ^{full} uses $\Lambda \le C_3^{(2)} \cdot 2 \cdot \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}})$, the reach-only assembly of Lemma 4.14. The withdrawn assembly carried the factor $1 + 2\sqrt{2} \approx 3.828$ [retracted 2026-09-04: p39-tensor-assembly-constant] instead and printed this column 1.914× larger. The supporting module's `propinquity_bound_full` field tracks $2\max(\gamma)$ *without* the $C_3^{(2)}$ factor (its 2 is documented there as a Lipschitz-comparison constant, a different quantity), so it agrees with this column only at $(2,2)$, where $C_3^{(2)} = 1$, and is not an independent corroboration of the factor 2 (corrected 2026-09-04; an earlier caption said the two "coincide"). Both columns vanish as $n_{\max,a}, n_{\max,b} \to \infty$; both exhibit monotone decrease across the diagonal panel.

Three structural observations.

1. *Monotone decrease and bit-identical match to single factor.* On the factorised panel, $\Lambda^{\text{fact}}(2,2) = 2.0746$, $\Lambda^{\text{fact}}(3,3) = 1.6101$, $\Lambda^{\text{fact}}(4,4) = 1.3223$. The diagonal ratio $\Lambda^{\text{fact}}(4,4)/\Lambda^{\text{fact}}(2,2) = 1.3223/2.0746 = 0.6374$ matches the single-factor Paper 38 ratio $\Lambda^{(1)}(4)/\Lambda^{(1)}(2) \approx 0.637$ *bit-identically* to four digits; the joint max-rate is dominated by the slower factor and inherits the single-factor cutoff scaling. On the full-operator-system column the analogous ratio is $\Lambda^{\text{full}}(4,4)/\Lambda^{\text{full}}(2,2) \approx 3.054/4.149 \approx 0.736$ (unchanged from the withdrawn column's 5.845/7.942, the factor cancelling), with the (now increasing) $C_3^{(2)}$ factor partially offsetting the $\max(\gamma)$ decrease.

2. *Joint cb-norm decreases as $1/(n_{\max,a} n_{\max,b})$.* $\|T^{(2,2)}\|_{\text{cb}} = 4/9 \approx 0.444$ vs. $\|T^{(4,4)}\|_{\text{cb}} = 4/25 = 0.16$, an exact factor of $9/25 = 0.36$ reduction matching the closed form of Lemma 4.3(c). The asymptotic order $O(1/(n_{\max,a} n_{\max,b}))$ is one order faster than the single-factor cb-norm $O(1/n_{\max})$.

3. *Test count.* The supporting computational module `geovac/gh_convergence_tensor.py` ($\sim$ 1881 lines) and the test file `tests/test_gh_convergence_tensor.py` contain the full verification panel, including `tests/test_paper39_triangle_tight.py`, which builds the joint commutator from the real Camporesi–Higuchi matrices and the graded multipliers and confirms (i) the triangle bound (17) is tight (ratio $\to 1$) and (ii) the Pythagorean form is maximally violated (ratio $\to 2$), the computational basis for the $C_3^{(2)} \le \sqrt{2}$ constant.

6 Related work and significance

6.1 Place in the Connes–van Suijlekom / Latrémolière lineage

The published propinquity tensor-product literature, as discussed in §1, is summarized in the following table.

Setup	Status	Reference
Two finite-dimensional	Standard	Connes–Chamseddine AC
Riemannian $\times$ finite	Closed	Latrémolière [14]
$C(X) \times A$ (A AF)	Closed	Aguilar [1]
Riemannian $\times$ abelian	Closed	Farsi–Latrémolière [7]
Single Riemannian along path	Closed	Farsi–Latrémolière [8]
Single truncation, NC integral	Closed	Hekkelman–McDonald [10]
Two infinite Riemannian non-flat	**Open prior to this work**	—
Two S^3 at distinct λ	**Closed (this paper)**	Theorem 3.1

The structural difficulty of the two-infinite case is that Latrémolière tunneling pairs do not multiply trivially: for $(B_a \otimes B_b, P_a \otimes P_b)$, the joint reach inherits from both single-factor approximate-identity rates with their commutator structure, and the joint Lipschitz constant is not $\max(C_3^a, C_3^b)$ in general. The Bożejko–Fendler product cb-norm formula covers each factor's central convolution but the propinquity-tunneling assembly for the product is not in print prior to this work. The present paper closes the case via the lifted-state construction (Lemma 4.12), which metrizes both state spaces by the translation seminorm and needs no dual reach. The (B, P)-pair route — using the Connes–Marcolli graded anticommutation (11) on the KO-dim-6 module, with joint commutator controlled by the triangle bound at constant $C_3^{(2)} \le \sqrt{2}$ (Lemmas 4.6, 4.14) — is retained as the abandoned approach.

6.2 Obstructions to the cross-manifold extensions

We documented in §1 that the same-manifold case treated here is the most accessible of three plausible tensor-product extensions of [20]. The two cross-manifold variants remain open structural questions in the broader noncommutative geometry program.

Riemannian $\times$ Hardy-sector ($\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\mathrm{Bargmann}}$). The first cross-manifold extension would tensor the Riemannian S^3 Camporesi–Higuchi triple with a Bargmann–Segal-type spectral triple on S^5 in the holomorphic Hardy sector [19]. This is structurally obstructed at the framework level: the S^3 factor is a second-order Riemannian spinor Dirac with KO-dimension 3, while the S^5 Bargmann factor is a first-order complex-analytic Euler operator on a Hardy sector and does not fit standard NCG Dirac axioms (KO-dimension is not classically defined in this setting). The two factors live in different categories of NCG construction, and there is no published prescription for tensoring them. We have flagged this elsewhere as the four-layer Coulomb/HO asymmetry of [19] §V resurfacing at the spectral-triple level. Neither the Bargmann transform nor the Kustaanheimo–Stiefel transformation provides a bridge: the former is a unitary intertwiner

of operators on Hilbert spaces, not a unitary equivalence of spectral triples, and the latter operates at the coordinate level, mapping the 3D Coulomb problem to a 4D HO problem under a constraint, without preserving the Riemannian / spinor bundle structure required for KO-dimension. A proper Riemannian–Hardy propinquity theory is not in the literature and would require new framework development beyond the scope of the present paper.

Riemannian × Riemannian on different manifolds ($\mathcal{T}_{S^3} \otimes \mathcal{T}^{\mathrm{Riem}}_{S^5}$). The alternative is to abandon the Bargmann content of S^5 and tensor S^3 with the round Riemannian S^5 spinor Dirac. KO-arithmetic is clean ($3+5=8\equiv 0 \pmod 8$), and this object is a textbook Connes–Marcolli construction. But propinquity convergence for $\mathcal{T}^{\mathrm{Riem}}_{S^5}$ alone — the rank-2 Spin(6) analog of [20] — requires a fresh five-lemma analysis: a higher-rank version of the Avery 3-Y integral, the corresponding SO(6) selection rule, and a rank-2 central spectral Fejér kernel. We expect analogous $\log n/n$-rate convergence with rank-dependent constants, but the detailed analysis is open and is itself a separate paper.

6.3 Significance for the multi-focal-composition program

The same-manifold case is what is needed for the GeoVac multi-focal composition program (where two electrons in distinct hydrogenic shells live on S^3 at distinct focal lengths $\lambda = n/Z$). With Theorem 3.1 in place, the cross-register two-body coordinate operator $V(\hat{r}_1, \hat{r}_2)$ across two electronic registers at distinct focal lengths is well-defined as a multiplier on the joint operator system $\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}$, with Lipschitz seminorm bounded by the product of $C_3^{(2)}(n_{\max,a}, n_{\max,b})$ and the classical Lipschitz norm of V on $S^3 \times S^3$. The convergence theorem provides the metric foundation; the structural application is left for a separate paper.

7 Open questions

(i) Quantitative rate at small $n_{\max}$. Theorem 3.1 inherits the single-factor asymptotic rate $\gamma_n \sim (4/\pi)\log n/n$ from [20]; the quantitative rate at small $n_{\max}$ remains the open follow-up to [20, §5(i)], parallel to single-factor Track C in the GeoVac internal record. The joint rate inherits the same open-quantitative-constant feature, since the max-form $\max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}})$ is dominated by the worse single-factor rate.

(ii) Off-factorized-panel functions. Theorem 3.1 is stated and verified on simple tensors $f \otimes g$ and extended by linearity to finite linear combinations; the $C_3^{(2)}$ formula (19) arises from the (tight) triangle bound on the graded Leibniz rule for unit-Lipschitz Avery harmonics, and is sharp on factorized harmonics ($C_3^{(2)} = \sqrt{2}$ is achieved). General $h \in C^\infty(S^3 \times S^3)$ that are not of factorized form may admit sharper bounds via direct Sobolev embedding on $S^3 \times S^3$. The detailed analysis of the off-factorized case is open.

(iii) Higher tensor factors — structural extension (sketch; the $k=2$ case is the theorem of this paper). The proof extends mechanically to k-fold tensor products $\mathcal{T}^{\lambda_1}_{S^3} \otimes \cdots \otimes \mathcal{T}^{\lambda_k}_{S^3}$ via repeated application of the graded Leibniz rule and the operator-norm triangle bound:

$$\Lambda^{(k)}\big(\mathcal{T}^{(k)}_{n_1,\ldots,n_k},\ \mathcal{T}^{(k)}_{(S^3)^{\otimes k}}\big) \ \le\ C_3^{(k)}(n_1,\ldots,n_k)\cdot \gamma^{(k)}_{n_1,\ldots,n_k}, \tag{30}$$

with the triangle bound

$$C_3^{(k)}(n_1,\ldots,n_k) \ \le\ \sqrt{k}\,\sqrt{\frac{N_*-1}{N_*+1}} \ \nearrow\ \sqrt{k}, \qquad N_* := \max_j n_j, \tag{31}$$

and joint rate $\gamma^{(k)} = \max_j \gamma_{n_j}^{\mathrm{SU}(2)}$ (worst-factor SU(2) rate). *Correction (2026-06-18):* an earlier draft claimed a k-**independent** constant $\sqrt{(N_*-1)/(N_*+1)} \to 1^-$ via a "Pythagorean Leibniz" sum-of-squares bound $\|\sum_j X_j\|_{\mathrm{op}}^2 = \sum_j \|X_j\|_{\mathrm{op}}^2$. That identity is **false**: pairwise anti-commutation of the $D^{(k)}$ summands does not make the resulting commutator terms operator-norm orthogonal (Remark 4.5; verified maximally violated at $k = 2$). The correct bound is the triangle one, which gives a genuine $\sqrt{k}$ factor (not eliminated, as had been claimed) from $\big(\sum_j (N_j - 1)\big)^2 \le k \sum_j (N_j - 1)^2$, hence $C_3^{(k)} \nearrow \sqrt{k}$. For each *fixed* k this is a finite constant, so convergence (30) still holds (driven by $\gamma^{(k)} \to 0$); but the constant grows with the number of factors and is *not* a k-independent drop-in tool. This $k > 2$ extension is a sketch: only the $k = 2$ case is proved and verified in this paper.

(iv) Conditional Coulomb/HO bridge. Should a unifying NCG-framework extension be developed that bridges the Riemannian spinor Dirac and Hardy-sector first-order complex operator categories at the spectral-triple level (§6.2), the W2b-medium case $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\mathrm{Bargmann}}$ would become a target for a follow-up to the present paper. The ingredients on the S^3 side are exactly Theorem 3.1 restricted to the left tensor factor.

(v) Tensor products at higher rank — structural extension (sketch). The present paper treats the equal-manifold tensor product $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ at two distinct focal lengths on the same compact group SU(2). A companion higher-rank generalisation of the single-factor result of [20] has since been established in [21], which lifts Theorem 3.1 of [20] to compact connected semisimple Lie groups G with bi-invariant metric (a circle factor carries the circle constant) and reports the rate constant $4/\pi$ (dual-Coxeter normalisation of the SU(2) metric; $2/\pi$ on the unit round S^3, the constant scaling with the metric — Paper 38, 2026-09-03) as *universal* across that class via a Plancherel-weight × Vandermonde-Jacobian cancellation (rigorous at rank 1).

Combining the k-fold extension (item (iii) above) with the general-G ingredients of [21] gives the expected multi-factor form

$$\Lambda^{(k)}\big(\mathcal{T}^{(k)}_{n_1,\dots,n_k},\ \mathcal{T}^{(k)}_{G_1\times\cdots\times G_k}\big) \ \le\ C_3^{(k)} \cdot \max_j \gamma_{n_j}^{G_j}, \qquad C_3^{(k)} \ \le\ \sqrt{k}\, \sup_j C_3^{G_j}(n_j) \ \nearrow\ \sqrt{k}, \tag{32}$$

for any $k \ge 1$ and arbitrary compact connected Lie groups $G_1, \dots, G_k$ with bi-invariant metrics, where $C_3^{G_j}(n_j) \to 1^-$ is the Paper 40 [21] per-factor L3 constant and $\gamma_{n_j}^{G_j}$ is the per-factor general-G rate at the universal $(4/\pi)\log n_j/n_j$ asymptote. *Correction (2026-06-18):* an earlier draft boxed this as a closed "master theorem" with a k-**independent** constant $\sup_j C_3^{G_j}(n_j)$; that rested on the withdrawn Pythagorean sum-of-squares bound (item (iii)). With the correct triangle bound the constant carries a genuine $\sqrt{k}$ factor and is *not* k-independent. For each fixed k convergence holds (finite constant), and the joint rate still inherits the universal $4/\pi$ asymptotic from [21]; but this multi-factor / general-G form is a **sketch** (only the $k = 2$, SU(2) case is proved here, and the $k = 1$ cases in [20, 21]), not a separately verified master theorem.

Practical applications: Paper 45 [22]'s compact-temporal carrier $S^3 \times S^1$ instantiates the $k = 2$, $G_1 = \mathrm{SU}(2)$, $G_2 = U(1)$ case of (32) at the level of its signature-agnostic product-carrier rate. Note, however, that the Lorentzian (Krein) reading of that carrier does *not* yield propinquity convergence: Paper 45 proves a degeneracy theorem (the natural Lorentzian-propinquity seminorm vanishes identically), so the master construction provides the surviving Riemannian product-carrier rate, not a Lorentzian convergence result.

8 Conclusion

We have proved — unconditional in the translation-seminorm metrization, the tier of Paper 38's `thm:main_unconditional` — van Suijlekom state-space Gromov–Hausdorff convergence

for truncations of the tensor-product Camporesi–Higuchi spectral triple on $S^3 \times S^3$ at distinct focal lengths (the strictly stronger Latrémolière propinquity is not claimed; see the convergence convention of §2), with explicit rate $\gamma_\lambda^{(ab)} \leq \lambda_a \gamma_{n_{\max,a}} + \lambda_b \gamma_{n_{\max,b}}$ of leading constant 1, via the lifted-state construction (Lemma 4.12), for which the dual reach is unnecessary. The earlier (B, P)-pair route (constant 2, joint Lipschitz comparison constant $C_3^{(2)} \leq \sqrt{2}$, dual-reach named gap) is retained as the record of an abandoned approach. In that abandoned route the mechanism controlling both the Lipschitz constant and the panel term was the Connes–Marcolli graded anticommutation $\{D_{\mathrm{CH}}^{(a)} \otimes I, \gamma_a \otimes D_{\mathrm{CH}}^{(b)}\} = 0$ — which is *vacuous as printed*, γ_a being necessarily 0 on S^3 (the note after (8)); on the KO-dimension-6 graded module (the Clifford-doubled operator, Lemma 4.13) it produces a clean two-term graded Leibniz rule, whose joint commutator is controlled by the (tight) operator-norm triangle bound $\|A + B\| \leq \|A\| + \|B\|$ (an earlier Pythagorean refinement is withdrawn — see Remark 4.5 — and convergence is unaffected since $C_3^{(2)} \leq \sqrt{2}$ is a finite constant). The result extends Connes–van Suijlekom [6] and the subsequent flat-structure and single-non-abelian-factor literature [9, 10, 15, 20] to the simplest non-abelian tensor case, and closes a case in the propinquity tensor-product programme that the published literature does not previously cover. The cross-manifold extensions remain open structural questions, with the Coulomb/HO category mismatch documented in [19] §V identifying a precise framework-level obstruction to the Riemannian–Hardy variant.

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[21] J. Loutey, *State-space Gromov–Hausdorff convergence of spectral truncations of compact semisimple Lie groups with bi-invariant metric*, preprint, May 2026; higher-rank single-factor companion to the present tensor-product paper, DOI via Zenodo. Generalizes [20] from SU(2) to all compact connected Lie groups; rate constant $4/\pi$ rigorous at rank 1 and numerically established at rank ≥ 2 (a rank-uniform symbolic proof is a named gap); asymptotic-tight $C_3 = 1$ via the PRV-summand bound of Kumar (1988) and Vinberg (1990).

[22] J. Loutey, *Lorentzian propinquity on truncated* $\mathrm{SU}(2) \times U(1)_T$ *Krein spectral triples: a degeneracy theorem*, preprint, May 2026; proves the natural Lorentzian-propinquity seminorm degenerates, closing that route; its signature-agnostic product-carrier rate is the $k = 2$, $G_1 = \mathrm{SU}(2)$, $G_2 = U(1)$ instance of the multi-factor form (32) (proved here only at $k = 2$; the general-G statement is a sketch, §7 (v)).